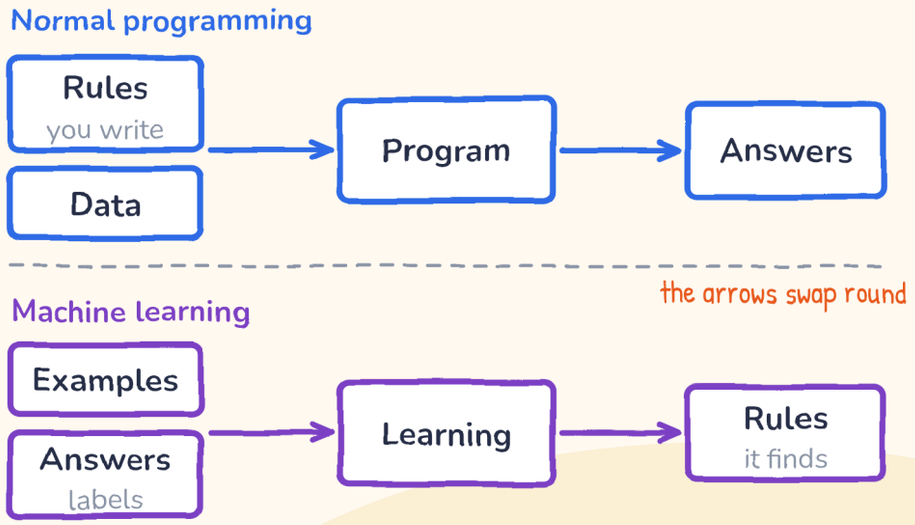
Artificial Intelligence · Machine Learning Foundations

What Is AI?

Write rules to spot a cat.
Now write them for every cat on Earth.

60 minutes | Pairs | Cards + a laptop



Did you know?

The words "artificial intelligence" were first used in 1956, for a summer workshop at Dartmouth College in the USA. It took about 60 more years for AI to fit in your pocket.

LEARN

Let's think

Write the rules for a cat

Try it in 60 seconds

Write rules a computer could follow to decide **cat or not a cat**.

Most people write something like: **fur, four legs, whiskers, pointy ears, says meow**.

Now break your rules

- A **Sphynx cat** has no fur at all.
- A **Scottish Fold** has folded-down ears.
- A cat curled into a ball shows **no legs**.
- A **dog** has fur, four legs and whiskers too.

Guess first

- How many rules would cover every cat on Earth?
- How did **you** learn what a cat is? Did anyone give you rules?
- Your phone can find every cat photo in your gallery. Who wrote its rules?

The big idea of today

You learned "cat" from seeing lots of cats, not from a list of rules. Modern AI learns the same way.

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Learning aims

Today I will learn to...

1  Tell a **rule-based program** from a **learning system**

2  Explain that AI **learns patterns from examples**

3  Find **AI in everyday apps** and say what it learned from

4  Show **where rules break** and examples help

Keywords

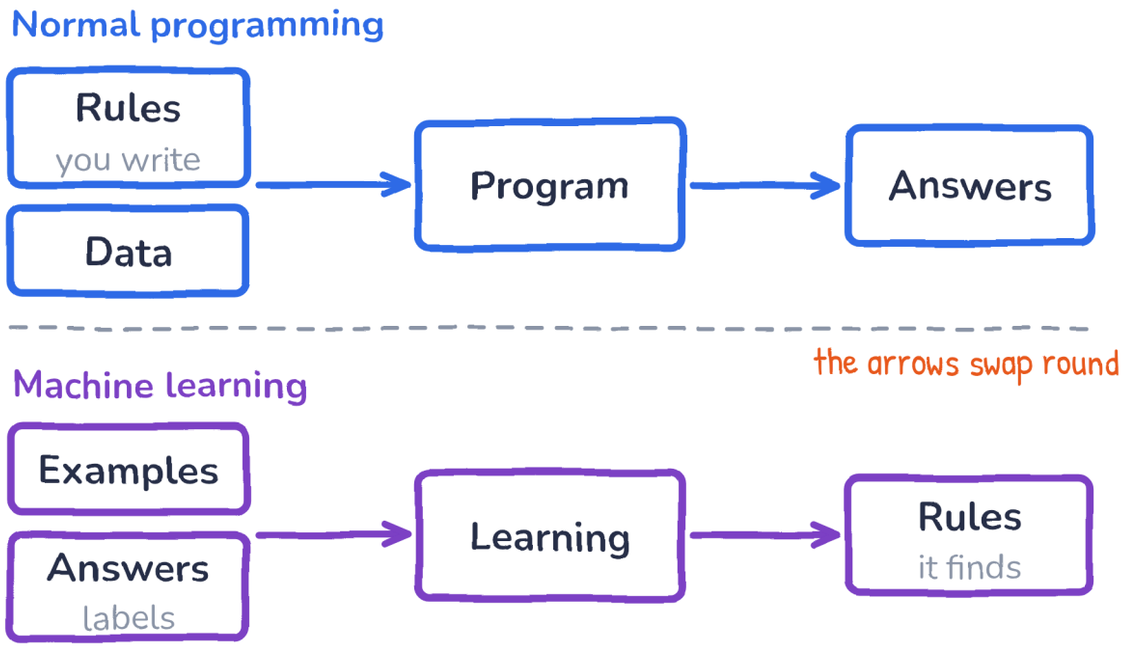
artificial intelligence · rule · program · example · label · pattern · training · prediction · confidence

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Objective 1 · The big idea

Two ways to make a computer smart

Rules in, or rules out



Normal programming

A person thinks of every case and writes it as a rule. The computer follows the rules exactly, every time.

Machine learning

A person collects **examples with the right answers**. The computer finds the pattern itself, and uses it on things it has never seen.

Artificial intelligence

Any computer doing something we would call smart if a person did it. Today almost all of it is machine learning.

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Objective 1 · Rules

A rule program, and the dog that fools it

```
# a cat detector made of rules
def is_cat(animal):
    if (animal["fur"]
            and animal["whiskers"]
            and animal["legs"] == 4):
        return "cat"
    return "not a cat"

dog = {"fur": True, "whiskers": True, "legs": 4}
print(is_cat(dog))   # cat  (wrong!)
```

Fix it with another rule?

Add "says meow". Now a photo can't be checked at all, because photos don't make sounds. Every fix makes a new problem.

Rules are the right tool when...

- every case is known: a **calculator**, a **Metro fare** machine
- the answer must be exact and explainable: **exam marks**, a **traffic-light timer**

Learning is the right tool when...

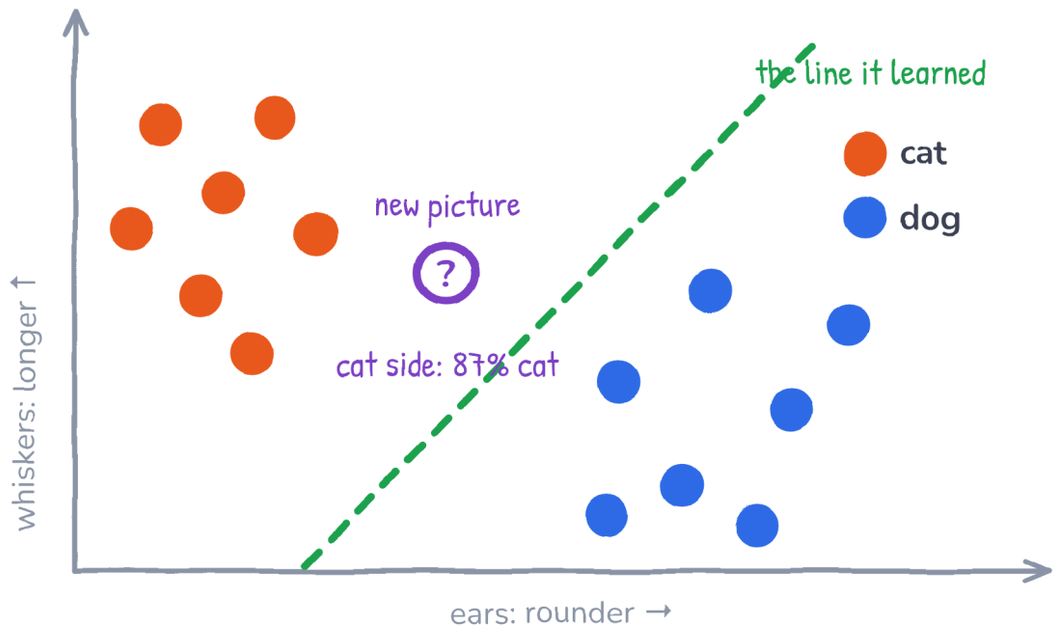
- there are too many cases to write: **faces**, **voices**, **handwriting**
- even experts can't say the rule: **is this email spam?**

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Objective 2 · Patterns

How a computer learns from examples

Every example is a dot with its answer



1. Examples with labels

Hundreds of pictures, each with the **right answer** written on it by a person: cat, or dog.

2. Training finds the pattern

The computer measures things (ear shape, whisker length) and finds the **line that best separates** the two groups.

3. A guess for something new

A new picture lands on one side of the line. That is the **prediction**, with how sure it is: 87% cat.

LEARN

Objective 2 · Patterns

Clever, but it doesn't understand



It only knows its examples

Train it only on orange cats and it may decide **orange = cat**. It learns whatever pattern is easiest, not what you meant.



It is always guessing

Even at 99% sure it can be wrong. A good AI shows how **confident** it is, so a person can check the unsure ones.



Better examples, better AI

More examples, and more **different** examples (other colours, angles, lighting), beat a cleverer computer.

A story from real life

A model trained to spot wolves in photos was very accurate, until people looked closer. It had learned that wolf photos had **snow** in them. Show it a husky in snow: "wolf".

Objective 3 · Everyday AI

AI is already in your pocket

Face unlock

Learned from millions of faces what makes **your** face yours, in any light.

Voice typing

Gboard types Urdu and English you speak. It learned from huge numbers of voice recordings.

Maps traffic

Google Maps knows the Islamabad Expressway is slow from the speed of phones already on it.

Recommendations

YouTube and Daraz suggest the next video or product from what people like you clicked.

Spam filters

Learned from millions of emails that people marked as spam or not spam.

Photo search

Type "beach" in your gallery. Nobody wrote a beach rule: it learned from labelled photos.

Part 2 · Make it

Now let's be the AI



Sort smart from not-smart, learn a secret rule from examples, then train two real AIs.

35 minutes | Pairs | Cards + a laptop

MAKE

Class game 1

Smart or not smart?

How to play

1. Each pair gets the 12 cards from the sheet.
2. Sort them into two piles: **follows rules** or **learned from examples**.
3. For every card, say **why** in one sentence.
4. Swap with the next pair and check their piles.

Watch out for the tricky ones

An **automatic door** feels clever, but it follows one rule: something moved, so open. Being useful is not the same as learning.

The test

Could a person write down every rule it uses? Then it's a rule program. If nobody could, it probably learned from examples.

MAKE

Class game 2

Be the AI: the human classifier

How to play

1. One student is the **AI** and steps outside.
2. The class picks a secret rule, like **things with wheels**.
3. The AI sees 6 picture cards marked **YES** or **NO**.
4. Now 4 new cards, unmarked. The AI says YES or NO. Count the right answers.
5. Show 6 more, very different examples, and test again.

What you just felt

Nobody told the AI the rule. It **found a pattern** in the examples, and more varied examples made its guesses better. That is training.

Round two

Pick a sneaky rule, like things that are red. If every YES card was also a car, the AI learns cars. Now you have seen what bad training data does.

MAKE

Train a real AI · 1 of 2

AI for Oceans

Do this

1. Open **code.org/oceans** (free, no login needed).
2. Teach A.I. Bot: label each object **fish** or **not fish**. Do at least 30.
3. Press **Continue** and watch it clean the ocean. Count its mistakes.
4. Next level: teach it a new word, like **angry** fish, using only pictures.

Notice

You never wrote a rule. You only gave **examples with answers**, and the bot found the pattern itself.

Ask yourself

When the bot calls a fish "angry", what was it really looking at? Its idea of angry is only as good as your labels.

MAKE

Train a real AI · 2 of 2

Quick, Draw!

Do this

1. Open **quickdraw.withgoogle.com** and press **Let's draw!**
2. Six doodles, 20 seconds each. The AI guesses out loud as you draw.
3. At the end, click a doodle it **missed**.
4. Look at what other people drew for that word. Why did yours not match?

Where it learned

It learned from over **50 million doodles** drawn by people all over the world, each one labelled with the word they were asked to draw.

Notice

It recognises how most people draw a thing, not the thing itself. Draw a cat your own way and it may never guess.

TEST

Test like an engineer

Where do rules break?

The rule	Something that breaks it	Would examples do better?
A cat has fur	A Sphynx cat (no fur)	Yes: show photos of every kind of cat
A ripe mango is yellow	A ripe Sindhri with green patches	Yes: show ripe and unripe mangoes
Spam says "you have won"	A real prize email from school	Yes: learn from emails people marked
A traffic light is red for 60 s	Nothing: it never changes	No: here a rule is the right tool

Your turn

Write your own rule for "a ripe mango", then try to break it. Can your partner find a mango your rule gets wrong?

LEVEL UP

Finished early?

Go further

1 

Write **5 rules** for spotting a **ripe mango**, then find 3 that break them

Which rule broke first? Could you ever finish the list?

2 

In AI for Oceans, teach a **new word** with deliberately bad labels

What did the bot learn instead? That is bias, made on purpose.

3 

Find a word Quick, Draw! **always misses** when you draw it

Look at its examples. What does everyone else draw that you don't?

4 

Find two apps at home that say **recommended for you**

Guess what examples each one learned from.

Check yourself

Can I really do it? Prove it.

1 Tell a **rule-based program** from a **learning system**

Proof: I sorted all 12 cards and gave a reason for each

2 Explain that AI **learns patterns from examples**

Proof: I explained the human classifier game in my own words

3 Find **AI in everyday apps**

Proof: I named two apps and what they learned from

4 Show **where rules break**

Proof: my partner broke my mango rule, or I broke theirs

Rate yourself: red = not yet · amber = nearly · green = yes, and I can show someone

WRAP UP

Show what you know

What did we learn?

Five things to keep

- A rule program does exactly what a person wrote, nothing more.
- Machine learning finds the rules itself, from examples with answers.
- AI learns patterns, so it only knows what its examples showed it.
- It is always guessing; good AI says how sure it is.
- Rules for the exact jobs, learning for the messy ones.

Exit ticket

A fruit shop wants a program that spots **ripe mangoes**. Rules or learning? Give one reason.

Next class

AI-02: Training data. You will teach a computer to see, with your own webcam, in ten minutes.